CS19541 – COMPUTER NETWORKS LAB MANUAL

PRACTICAL 14

AIM: To write a program using RAW sockets to implement packet sniffing.

SOFTWARE REQUIRED:

Python 3 and a computer with a network interface. On Windows, raw packet capture may require additional tools and administrative privileges; this example is intended for Linux.

ALGORITHM:

1. Import the socket module.
2. Create a RAW socket using the appropriate protocol.
3. Bind the socket to a network interface if required by the operating system.
4. Receive packets continuously from the network.
5. Read the IP header and identify the source and destination addresses.
6. Display packet information such as protocol, source IP, destination IP, and packet length.
7. Stop the program using Ctrl+C.

PROGRAM:

```
import socket
import struct
import signal

def packet_sniffer():     # Linux: raw
socket receives IP packets     sniffer =
socket.socket(         socket.AF_PACKET,
socket.SOCK_RAW,          socket.ntohs(3)
)     print("Packet sniffing started.
Press Ctrl+C to stop.")

    try:         while True:
packet, address =
sniffer.recvfrom(65535)

            # Ethernet header is 14 bytes;
IPv4 header follows it              if
len(packet) < 34:                 continue

            eth_type = struct.unpack("!H",
packet[12:14])[0]             if eth_type
!= 0x0800:   # IPv4 only
continue

            ip_header = packet[14:34]
            iph =
struct.unpack("!BBHHHBBH4s4s", ip_header)
version_ihl = iph[0]             protocol
= iph[6]
```

```
            src_ip =
socket.inet_ntoa(iph[8])
dst_ip = socket.inet_ntoa(iph[9])


print("Protocol:", protocol,
"| Source:", src_ip,
                  "|
Destination:", dst_ip,
"| Length:", len(packet))
except KeyboardInterrupt:
        print("\nPacket
sniffing stopped.")
finally:
sniffer.close()

if __name__ == "__main__":
packet_sniffer()
```

PROCEDURE:

1. Save the program as **packet_sniffer.py**.
2. Open a Linux terminal in the folder containing the file.
3. Run the program with administrator privileges: **sudo python3 packet_sniffer.py**.
4. Generate network traffic by opening a website or using the ping command.
5. Observe the source IP, destination IP, protocol number, and packet length displayed in the terminal.
6. Press Ctrl+C to stop packet capture.

SAMPLE OUTPUT:

```
Packet sniffing started. Press Ctrl+C to stop.
Protocol: 6 | Source: 192.168.1.5 | Destination: 142.250.72.14 | Length: 74
Protocol: 17 | Source: 192.168.1.5 | Destination: 192.168.1.1 | Length: 86 Packet
sniffing stopped.
```

RESULT

Thus, a packet-sniffing program using RAW sockets was written and its operation was demonstrated by displaying captured packet details.